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Comparison of Traditional and Hybrid Time Series Models for Forecasting COVID-19 Cases

Prajapati, Samyak; Aman Swaraj; Lalwani, Ronak; Narwal, Akhil; Verma, Karan . Prajapati, Samyak; Aman Swaraj; Lalwani, Ronak; Narwal, Akhil; Verma, Karan.

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ABSTRACT (ENGLISH)

Time series forecasting methods play critical role in estimating the spread of an epidemic. The coronavirus outbreak of December 2019 has already infected millions all over the world and continues to spread on. Just when the curve of the outbreak had started to flatten, many countries have again started to witness a rise in cases which is now being referred as the 2nd wave of the pandemic. A thorough analysis of time-series forecasting models is therefore required to equip state authorities and health officials with immediate strategies for future times. This aims of the study are three-fold: (a) To model the overall trend of the spread; (b) To generate a short-term forecast of 10 days in countries with the highest incidence of confirmed cases (USA, India and Brazil); (c) To quantitatively determine the algorithm that is best suited for precise modelling of the linear and non-linear features of the time series. The comparison of forecasting models for the total cumulative cases of each country is carried out by comparing the reported data and the predicted value, and then ranking the algorithms (Prophet, Holt-Winters, LSTM, ARIMA, and ARIMA-NARNN) based on their RMSE, MAE and MAPE values. The hybrid combination of ARIMA and NARNN (Nonlinear Auto-Regression Neural Network) gave the best result among the selected models with a reduced RMSE, which proved to be almost 35.3% better than one of the most prevalent method of time-series prediction (ARIMA). The results demonstrated the efficacy of the hybrid implementation of the ARIMA-NARNN model over other forecasting methods such as Prophet, Holt Winters, LSTM, and the ARIMA model in encapsulating the linear as well as non-linear patterns of the epidemical datasets.

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